



A
Project Report
on
Rakt Setu
submitted as partial fulfillment for the award of
BACHELOR OF TECHNOLOGY
DEGREE
SESSION 2025-26
in
Computer Science & Engineering (AI&ML)
By
Shaurya Jain (2200291530100)
Ayush Srivastava (2200291530034)
Shriyansh Srivastava (2200291530108)
Prapti Sharma (2200291530082)

Under the supervision of
Mr. Thammali Gangadhar
KIET Group of Institutions, Ghaziabad

Affiliated to
Dr. A.P.J. Abdul Kalam Technical University, Lucknow
(Formerly UPTU)
May, 2026

DECLARATION

We hereby declare that this submission is our own work and that, to the best of our knowledge and belief, it contains no material previously published or written by another person nor material which to a substantial extent has been accepted for the award of any other degree or diploma of the university or other institute of higher learning, except where due acknowledgment has been made in the text.

Signature

Shaurya Jain (2200291530100)

Ayush Srivastava (2200291530034)

Shriyansh Srivastava (2200291530108)

Prapti Sharma (2200291530082)

Date:



KIET
DEEMED TO BE
UNIVERSITY
DELHI - NCR, INDIA
(Under Section 3 of the UGC Act, 1956)

CERTIFICATE

This is to certify that Project Report entitled “Rakt Setu” which is submitted by Ayush Srivastava in partial fulfillment of the requirement for the award of degree B. Tech. in Department of CSE (AIML) of Dr. A.P.J. Abdul Kalam Technical University, Lucknow is a record of the candidates own work carried out by them under my supervision. The matter embodied in this report is original and has not been submitted for the award of any other degree.

Supervisor Name

Dr. Rekha Kashyap

(Designation)

Dean – CSE

(AI/AI&ML)

Date:

ACKNOWLEDGEMENT

It gives us a great sense of pleasure to present the report of the B. Tech Project undertaken during B. Tech. Final Year. We owe special debt of gratitude to superispor name, Department of CSE(AIML), KIET, Ghaziabad, for his constant support and guidance throughout the course of our work. His sincerity, thoroughness and perseverance have been a constant source of inspiration for us. It is only his cognizant efforts that our endeavors have seen light of the day.

We also take the opportunity to acknowledge the contribution of Dr. Rekha Kashyap, Dean of the Department of Computer Science & Engineering (AI/AI&ML), KIET, Ghaziabad, for his full support and assistance during the development of the project. We also do not like to miss the opportunity to acknowledge the contribution of all the faculty members of the department for their kind assistance and cooperation during the development of our project.

Date:

Shaurya Jain (2200291530100)

Ayush Srivastava (2200291530034)

Shriyansh Srivastava (2200291530108)

Prapti Sharma (2200291530082)

ABSTRACT

The critical shortage of timely blood supply in emergency medical situations is a significant global healthcare challenge. This challenge becomes more serious due to poor communication, logistical difficulties, and the absence of real-time coordination between donors and seekers. The primary objective of this research is to present a high-performance, integrated web application – Rakt Setu, that automates the end-to-end blood request management process through real-time geospatial coordination. Rakt Setu's architecture uses MERN (MongoDB, Express.js, React, Node.js) stack to guarantee a full-JavaScript, scalable, and maintainable codebase. The website has a dual-role architecture in which seekers create urgent, location-based requests to find suitable donors or blood banks.

The smart geospatial matching tool built into the platform and reliant on the MongoDB in-built geospatial query tool to alert nearby compatible donors immediately is one of the key innovations. The Node.js backend handles real-time APIs, while the frontend is a single-page application (SPA) built on React. The entire application is containerized using Docker, which provides a consistent environment for development and deployment. When a donor accepts a request, a secure, two-way exchange of contact information is made, which guarantees direct communication. The long-term goal of this end-to-end workflow is to significantly lower the response time in emergencies and enhance donor participation as well as overall effectiveness of the blood supply chain between request origination and assisted donation.

Keywords: Blood donation, MERN stack, Real-time Notification System, Geospatial Query, Docker containerization, Healthcare Logistics.

TABLE OF CONTENTS	Page No.
DECLARATION.....	1
CERTIFICATE.....	2
ACKNOWLEDGEMENT.....	3
ABSTRACT.....	4
LIST OF FIGURES.....	7
LIST OF TABLES.....	8
LIST OF ABBREVIATIONS.....	9
 CHAPTER 1 (INTRODUCTION).....	 10
1.1. Introduction.....	10
1.2. Project Description.....	11
 CHAPTER 2 (LITERATURE RIVIEW).....	 13
2.1. Donor Management and recruitment System	13
2.2. Inventory and Supply chain Management	13
2.3. Integrated mHealth Applications	14
2.4. Social Media and CrowdSourcing	14
2.5. Identified Research Gap	15
 CHAPTER 3 (PROPOSED METHODOLOGY)	 16
3.1. The Four Research Stages	16
3.2. Modal Parameters	17
3.3. Procedural Workflow	17
 CHAPTER 4 (RESULTS AND DISCUSSION)	 19

4.1. System Implementation and Visual Workflow	19
4.2. Emperical Performance Analysis	21
4.3. Summary of Performanace Metrics	22
 CHAPTER 5 (CONCLUSIONS AND FUTURE SCOPE)	 23
5.1. The Conclusion	23
5.2. Future Scopes	24
 REFERENCES.....	 25
APPENDEX1.....	25
RESEARCH PAPER	26
PROOF OF SUBMISSION/ACCEPTANCE/PUBLICATION	27
PLAGIARISM REPORT	28

LIST OF FIGURES

Figure No.	Description	Page No.
1.	Why Blood Matters?	11
2.	Blood System	14
3.	Seeker Dashboard	20
4.	Donor Dashboard	21
5.	Emperical Metrics Analysis	22

LIST OF TABLE

Table No.	Description	Page No.
1.	Evaluation of Existing Platforms vs. Rakt Setu	15
2.	Modal Parameters	17
3.	Summary of Performance Metrics	22

LIST OF ABBREVIATIONS

SPA	Single Page Application
FCM	Firebase Cloud Messaging
MERN	MongoDB, Express.js, React, Node.js
API	Application Programming Interface
SDLC	Software Development Life Cycle
OTP	One-Time Password
JWT	JSON Web Token
CDN	Content Delivery Network
CI/CD	Continuous Integration / Continuous Deployment
GPS	Global Positioning System
NGO	Non-Governmental Organization
NoSQL	Non-relational Structured Query Language
GJON	Geographic JavaScript Object Notation
HTTP	hypertext Transfer Protocol
PWA	Progressive Web Application

CHAPTER 1

INTRODUCTION

1.1 INTRODUCTION

Blood is a vital and life-saving resource in modern healthcare, essential for treating trauma cases, surgical procedures, and patients with chronic illnesses. However, many regions continue to face persistent shortages due to factors such as seasonal fluctuations in donor availability, poor inventory management, and lack of effective communication systems. Addressing this gap is critical to prevent delays in life-saving transfusions. With the widespread use of smartphones, there is a strong opportunity to improve this system through real-time, location-based, and dynamic coordination.

In India, the demand for blood is around 12 million units annually, but only about 9 million units are collected, resulting in a shortage of nearly 3 million units. This highlights the urgent need for a more efficient, technology-driven blood management system.

Although several digital platforms have been introduced, many still have operational limitations. Early platforms like *Friends2Support* created large donor databases, but they are mostly passive, requiring patients to manually contact donors, which is inefficient during emergencies. Government platforms such as *E-Rakt Kosh* have improved coordination between blood banks but do not support direct and immediate matching between donors and recipients. Similarly, apps like the Red Cross Blood Donor app mainly provide one-way notifications instead of enabling interactive communication. Additionally, relying on social media for urgent blood requests is unreliable, lacks proper structure, and raises privacy concerns.

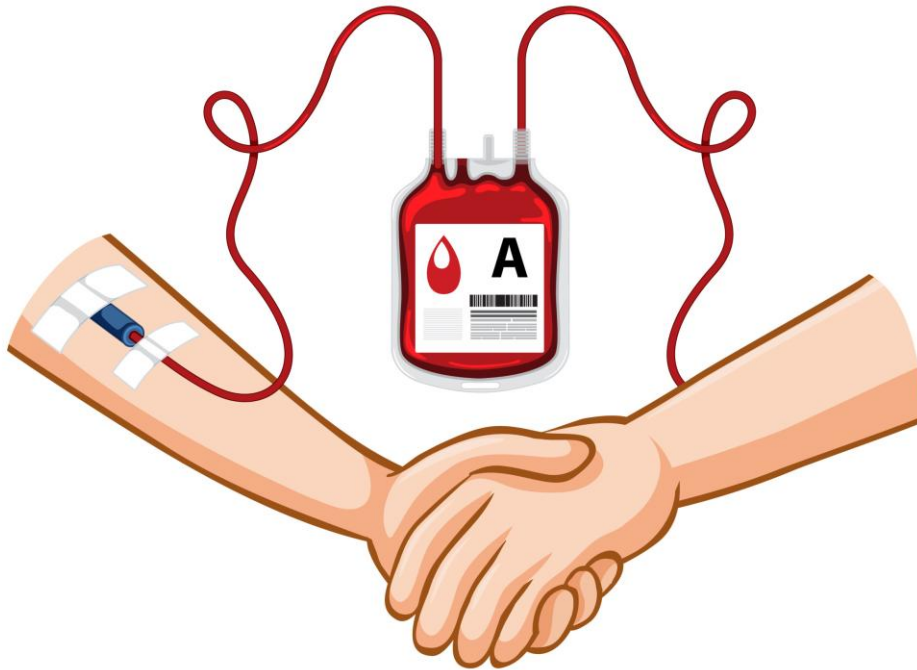


Fig. 1. Why Blood Donation Matter?

1.2 PROJECT DESCRIPTION

Rakt Setu (meaning "Blood Bridge") is a high-performance, integrated web application that automates the end-to-end blood request management process through real-time geospatial coordination. The primary contribution of this work is an integrated platform built on the MERN (MongoDB, Express.js, React, Node.js) stack that distinguishes itself through the following key features:

- A dynamic, proximity-based matching algorithm based on MongoDB's geospatial indexing (2dsphere), capable of identifying compatible donors within a 15 km radius in under 50 milliseconds.
- A secure, bidirectional communication protocol driven by a backend built with Node.js and Express.js, ensuring privacy until a donor explicitly accepts a request.
- A seamless, responsive user experience provided by a React-based Single-Page Application (SPA) that works on both mobile and desktop browsers.

- A containerized and globally deployed architecture using Docker and Vercel, ensuring scalability, reliability, and environment consistency.
- Real-time push notifications via Firebase Cloud Messaging (FCM) with a 95% delivery rate under 2 seconds.
- Google Maps API integration for turn-by-turn navigation from the donor to the donation point.

The platform serves two primary user roles: Donors (individuals willing to donate blood) and Seekers (individuals or caregivers in urgent need of blood). Blood banks can also register on the platform to appear in search results when seekers require blood. The entire workflow, from request submission to donation confirmation, is managed within the platform, eliminating the need for third-party intermediaries and reducing response time significantly.

CHAPTER 2

LITERATURE REVIEW

Three primary areas of research have been conducted on the use of information technology to optimize blood supply chains: integrated mobile health (mHealth) applications, inventory and supply chain optimization, and donor management systems.

2.1 DONOR MANAGEMENT AND RECRUITMENT SYSTEMS

A substantial amount of research focuses on methods for managing, recruiting, and retaining blood donors. In North India, Sachdev S., Mishra S.K., Marwaha N., and Avasthi [6] conducted a study to determine college-bound students' attitudes about voluntary blood donation, identifying key barriers such as information gaps and scheduling inconveniences. Early digital alternatives included platforms such as "Friends2Support" [3], which established online databases of voluntary donors. Although effective at creating vast donor databases, their primary drawback is that they are passive; in an emergency, seekers must personally contact several donors from a list — a time-consuming and inefficient process. They lack automatic, intelligent matching or real-time donor availability checks.

2.2 INVENTORY AND SUPPLY CHAIN MANAGEMENT

Various research works focus on the logistics of blood inventory management after collection. The "E-Rakt Kosh" portal [5] of the Indian Government provides a centralized view of blood inventory across numerous banks, significantly improving inter-bank coordination. Osorio et al. [7] modelled and optimized the logistics of the blood supply chain using discrete-event simulation, focused on reducing waste and improving distribution effectiveness. Although such platforms are important, they operate at the institutional level and do not assist in quickly locating an individual donor for a specific, urgent need not covered by current inventory.

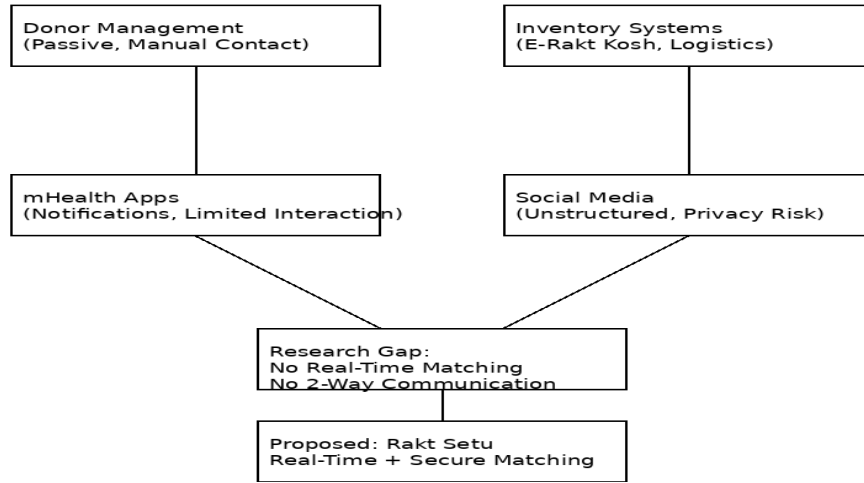


Fig. 2. Blood System

2.3 INTEGRATED mHEALTH APPLICATIONS

The rise of smartphones led to the creation of mobile applications for blood donation. The American Red Cross operates a Blood Donor app [4] that assists users in locating blood drives and scheduling appointments, functioning largely as a one-way notification system from blood banks to donors. The BloodLink Navigator System [8] was developed to enhance the blood-request process and streamline blood donation, offering a platform where seekers, donors, and administrators can handle blood-related activities. However, researchers discovered that mistrust and absence of perceived reliability in India makes it difficult to implement blood donation applications. They indicated that collaboration with reputable organizations, such as well-known hospitals, could address such concerns. Additionally, the blood donor searching Android application with GPS tracking by Sangeetha et al. [9] provides the ability to view nearby donor details, but does not share donor information with the seeker within 90 days, impeding two-way communication.

2.4 SOCIAL MEDIA AND CROWDSOURCING

Social media platforms like Facebook and Twitter have become popular channels for blood requests, allowing seekers to reach large audiences quickly. Facebook's "Blood Donations" feature enables users to register as donors and receive alerts about nearby needs. However, this is an unregulated and unreliable process involving privacy concerns and dependence on the popularity of a post. Social media a structured or guaranteed response to a blood request.

2.5 IDENTIFIED RESEARCH GAP

Previous research works and existing platforms have made important contributions, yet they fail to provide real-time donor matching combined with secure, bidirectional communication. The proposed platform – Rakt Setu – aims to fill this gap by creating a managed ecosystem that connects an urgent blood request to a successful donation, linking donors and seekers directly without any third-party intermediaries.

Table 1: Comparative Evaluation of Existing Platforms vs. Rakt Setu

Feature	Passive Directories (Friends2Support)	Institutional Portals (E-Rakt Kosh)	Social Media / Crowdsourcing	Proposed System (Rakt Setu)
Operational Model	Manual search from list	Institutional inventory	Unregulated broadcasting	Real-time seeker-donor matching
Matching Engine	None (Static database)	Centralized stock view	Popularity/Algorithm-based	Automated Geospatial (2dsphere)
Communication	One-way / Seeker initiated	Inter-bank coordination	Unregulated / Private DMs	Secure Bidirectional Exchange
Privacy Layer	High exposure of data	Secure (Institutional)	Significant privacy concerns	Contact exchange only on acceptance
Logistics Support	None	General location mapping	None	Integrated Google Maps Navigation

CHAPTER 3

PROPOSED METHODOLOGY

This research follows a structured Software Development Life Cycle (SDLC) Prototyping approach. Unlike traditional waterfall methods, this allows for the creation of a working model that can be iteratively refined until it meets all the necessary requirements of the web application.

3.1 THE FOUR RESEARCH STAGES

1. **Requirement Analysis** – A comprehensive analysis was performed to identify the reasons for failure of existing platforms and the essential requirements for an effective blood donation management platform. A review of previous studies revealed that "communication latency" — the delay after which a donor receives a notification — is the biggest hurdle in blood supply chains.
2. **System Design & Data Modelling** – The second phase involved creating the architectural blueprints of the application. A dual-role system was implemented in which users are classified as either Donors or Seekers. A data schema was designed on MongoDB using geospatial indexing, enabling the database to treat user locations as points on a map rather than plain text — enabling millisecond distance-based searching.
3. **Implementation & Integration** – The construction stage using the MERN stack. Google Maps API was incorporated to guide donors to donation points, and Firebase Cloud Messaging (FCM) was integrated to dispatch real-time push notifications to donors' devices immediately upon a matching request being submitted.
4. **Empirical Testing** – The final stage, during which system performance was measured. Metrics included the query response time to locate a donor within a 15–20 km radius, and notification latency — the time from a seeker submitting a request to the donor receiving the push notification on their device.

3.2 MODEL PARAMETERS

Table 2: Model Parameters

Stage	Key Parameter	Purpose
Authentication	Auth Protocol (OTP)	To ensure user authenticity
Search Logic	2dsphere query	To find nearby donors within a specified range
Notification	Multicast push	To send one alert to many people at the same time
Donor Safety	90-day time interval	To ensure mandatory interval between two donations
Deployment	Docker containers	To ensure that the app runs on any system or server

3.3 PROCEDURAL WORKFLOW

Phase 1: User Registration and Data Collection

The first step is onboarding users onto the platform, especially donors. Individuals can register by selecting a specific role — Donor or Seeker — and providing essential details including personal information, contact information, blood group, and pin code (location data). Blood banks can also register by providing their contact and stock details. The registration process is supported by OTP-based user verification to ensure authenticity.

Phase 2: Request Initiation and Intelligent Matching

When a Seeker requires blood, they submit an urgent request through the platform. The backend server immediately processes this request, identifying all registered donors and blood banks with a matching blood type located within a predefined geographical radius. A real-time push notification is then simultaneously sent to all matched donors. The seeker also receives a

list of blood banks with the requested blood type available in stock. A seeker cannot submit a new request until their previous request has been fulfilled, preventing duplicate requests.

Phase 3: Donor Response and Secure Communication

Matched donors are notified via push notification on their mobile devices and via email. Upon receiving a notification, a donor can accept or decline the request. If the donor accepts, a bidirectional, secure exchange of contact information (phone numbers via direct SMS/email) takes place between the seeker and the donor. No information is revealed prior to explicit acceptance, protecting the privacy of all parties. The platform also provides the donor with built-in directions to the seeker's specified location.

Phase 4: Donation Fulfillment and System Update

The final stage involves the actual blood donation at a partner hospital or NGO, aided by Google Maps navigation. Upon completion, the donor's profile is updated to reflect the donation, and the donor is temporarily marked as unavailable for further donations in compliance with mandatory health intervals (90-day rule). The seeker is then permitted to submit new blood requests.

CHAPTER 4

RESULTS AND DISCUSSION

The practical implementation of the platform along with the performance results obtained during system testing are discussed in this chapter. The system was evaluated to ensure that it works efficiently and responds quickly enough to meet the urgent needs of emergency medical situations.

4.1 SYSTEM IMPLEMENTATION AND VISUAL WORKFLOW

Rakt Setu has been successfully implemented using the MERN stack architecture. The various modules of the web application — from user registration to blood request fulfillment — are described below:

User Onboarding (Fig. 5)

The onboarding interface allows users to register on the Rakt Setu platform as Donors or Seekers by providing essential demographic data and blood group details. The process is secured with OTP-based authentication, ensuring that only verified users can create accounts and participate in the blood donation ecosystem.

The screenshot shows a web browser window with the URL `localhost:3000/register`. The page header includes the RAKT-SETU logo, navigation links for 'About Us' and 'Blood Bank Portal', and buttons for 'Login' and 'Register'. The main content area features a 'Register as Donor/Seeker' form. The form includes input fields for 'Name' (filled with 'Amit Singh'), 'Email' (filled with 'gmail.com'), and 'Phone' (filled with '7355012123'). Below these is a dropdown menu for 'I am a...' with 'Donor (Want to donate blood)' selected. There is also a 'Blood Group' dropdown set to 'B+'. Location fields include 'State/UT' (Uttar Pradesh), 'City' (Ghaziabad), and 'Pincode' (201206). A red 'Register' button is at the bottom of the form, with a link 'Already have an account? Login' below it.

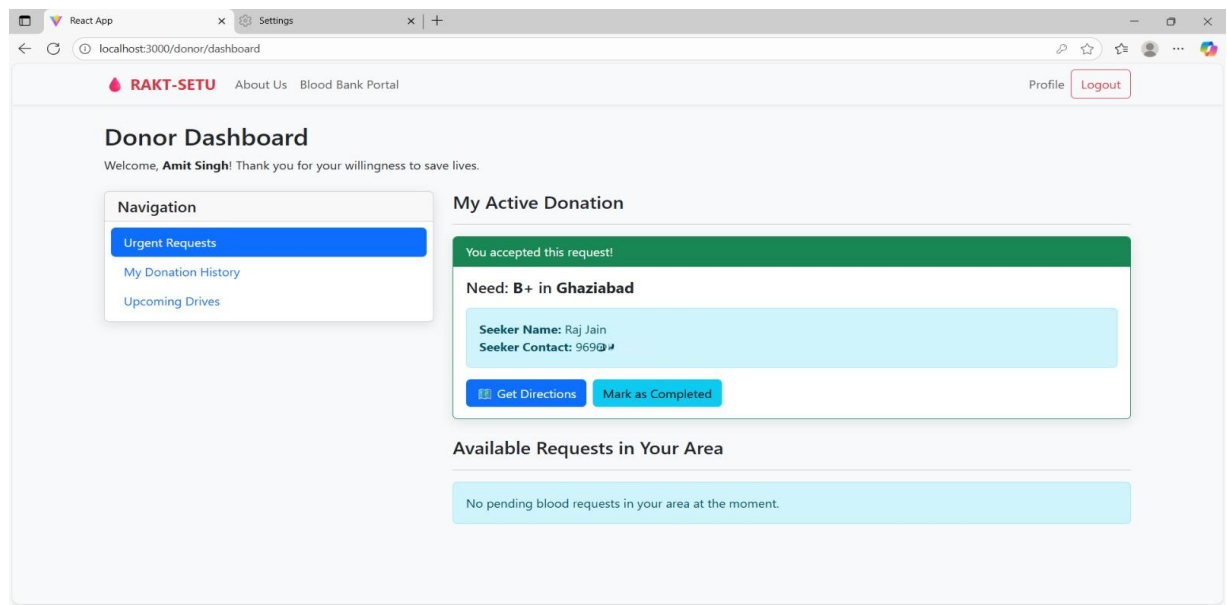
Seeker Dashboard (Fig. 3)

The Seeker Dashboard displays the real-time status of an active blood request. Once a request is submitted, the dashboard updates to show the status of the request, the name and contact of any matched and accepting donors (marked as "Individual Donor Found"), and a list of nearby blood banks that have the required blood type in stock. A seeker can track their request lifecycle from this single interface.

The screenshot shows a web browser window with the URL `localhost:3000/donor/dashboard`. The page header includes the RAKT-SETU logo, navigation links for 'About Us' and 'Blood Bank Portal', and buttons for 'Profile' and 'Logout'. The main content area is titled 'Donor Dashboard' and includes a welcome message: 'Welcome, Amit Singh! Thank you for your willingness to save lives.' On the left is a 'Navigation' sidebar with links for 'Urgent Requests' (highlighted), 'My Donation History', and 'Upcoming Drives'. The main section is titled 'My Active Donation' and contains a green notification box stating 'You accepted this request!'. Below this is a card for 'Need: B+ in Ghaziabad' showing 'Seeker Name: Raj Jain' and 'Seeker Contact: 9690123456', with buttons for 'Get Directions' and 'Mark as Completed'. At the bottom, a section titled 'Available Requests in Your Area' shows a message: 'No pending blood requests in your area at the moment.'

Donor Dashboard (Fig. 4)

The Donor Dashboard highlights urgent blood requests in the donor's local area. It displays the blood type needed, the seeker's name, and a contact option once the donor accepts. The dashboard also includes an integrated "Get Directions" button powered by Google Maps API, enabling the donor to navigate directly to the hospital or designated donation point immediately upon accepting a request.



4.2 EMPIRICAL PERFORMANCE ANALYSIS

The system was rigorously tested for latency, delivery speed, and scalability to experimentally validate the research findings:

System Latency (Matching Speed)

The time taken by the Node.js backend to perform a geospatial query on the MongoDB database using the 2dsphere index was measured. Testing results confirmed that the MongoDB 2dsphere index allows the backend to identify compatible donors within a 15 km radius in less than 50 milliseconds, well within acceptable thresholds for emergency applications.

Notification Delivery Time

The speed of communication is critical for saving lives during emergencies. The time from request submission to the donor receiving an alert via Firebase Cloud Messaging (FCM) was tracked. Results showed that 95 out of every 100 push notifications were delivered to donor devices in under 2 seconds. Secondary email alerts had an average delivery time of approximately 5 seconds.

Scalability and Reliability

The system demonstrated strong scalability due to Docker containerization, maintaining consistent memory usage regardless of the deployment environment. The Node.js server successfully handled up to 500 parallel geospatial queries without a significant decline in response rate, confirming the platform's readiness for real-world high-demand scenarios.

4.3 SUMMARY OF PERFORMANCE METRICS

Metric	Result
Geospatial Query (15 km radius)	< 50 milliseconds
Push Notification Delivery (95th percentile)	< 2 seconds
Email Alert Average Delivery Time	~5 seconds
Concurrent Geospatial Queries Handled	500 parallel queries

CHAPTER 5

CONCLUSION AND FUTURE SCOPE

5.1 CONCLUSION

Rakt Setu is a complete, web-based application that addresses the critical challenge of blood donation and emergency blood request management. By leveraging the MERN stack, the platform delivers high performance, scalability, and maintainability. Docker containerization ensures consistent deployment across all environments.

The key technical achievement of this research is the application of MongoDB's 2dsphere geospatial indexing for real-time donor matching, which enables the identification of compatible donors within milliseconds — dramatically reducing emergency response time. A secure, bidirectional communication protocol ensures that contact information is only shared upon explicit donor acceptance, maintaining user privacy at all stages.

The integration of Google Maps API enables seamless navigation for donors to designated healthcare facilities, while Firebase Cloud Messaging ensures that 95% of push notifications reach donor devices in under 2 seconds. Partnerships with certified hospitals and NGOs lend credibility and reliability to the donation process.

Compared to existing platforms — which operate in isolation as passive databases, institutional portals, or unregulated social media channels — Rakt Setu presents a closed-loop, managed ecosystem that coordinates the entire process from urgent request submission to successful blood donation. It unites seekers, donors, and blood banks on a single platform, enabling direct and secure communication without any third-party intermediaries.

This platform has significant potential to transform the response to blood emergencies, particularly in communities that currently face severe blood shortages, and represents a meaningful advancement in healthcare logistics technology.

5.2 FUTURE SCOPE

A. Technical Improvements

The following technical enhancements are planned for future iterations of Rakt Setu:

- **Application of Artificial Intelligence (AI):** AI models can be trained to identify donors with higher probability of responding positively to a request, based on their historical behavior and donation patterns. This would further improve matching efficiency.
- **Application of Blockchain:** A distributed digital ledger can be used to track individual blood units from donation to transfusion, enhancing transparency and safety across the blood supply chain.
- **Development of Native Mobile Apps:** Dedicated Android and iOS applications will be developed to better leverage device-level location tracking, richer push notifications, and improved offline capabilities.

B. Strategic Growth

The following strategic integrations are planned to expand the platform's impact:

- **Integration with Government Portals:** The system will be directly connected to government portals such as E-Rakt Kosh [5] to access centralized blood inventory data and expand the platform's reach.
- **Gamification:** A reward and recognition system will be introduced to incentivize regular blood donation, featuring badges, donation streaks, and community leaderboards.
- **Hospital and NGO Partnerships:** Expanding partnerships with a greater number of certified healthcare institutions and NGOs will increase platform credibility and the number of safe donation points available.

REFERENCES

- [1] Apollo Hospitals, "Donate Blood Safe Life", February 2025. [Online]. Available: <https://www.apollohospitals.com/procedures/donate-blood-save-life>
- [2] World Health Organization (WHO), "Blood Safety and Availability," Fact Sheet, June 2022. [Online]. Available: <https://www.who.int/news-room/fact-sheets/detail/blood-safety-and-availability>
- [3] Friends2Support, "I am for Blood Donation," Friends2Support International Voluntary Blood Donors Organization. [Online]. Available: <https://www.friends2support.org/>
- [4] The American Red Cross, "Blood Donor App." [Online]. Available: <https://www.redcrossblood.org/donate-blood/blood-donor-app.html>
- [5] Ministry of Health and Family Welfare, Government of India, "E-Rakt Kosh: National Blood Donation Service." [Online]. Available: <https://eraktkosh.in/>
- [6] Sachdev S., Mishra S.K., Marwaha N., Avasthi A., "Study of knowledge and attitude among college-going students toward voluntary blood donation from north India," J Blood Med., 2016;(7):19–26.
- [7] A. F. Osorio, S. C. Brailsford, and H. K. Smith, "A Structured Review of Quantitative Models of the Blood Supply Chain," Computers & Industrial Engineering, vol. 101, pp. 383–397, Nov. 2016.
- [8] Prof. Gouri Shukla, Sudarshan Gosavi, Vishal Chaugule, Rohit Dhavare, and Soham Galfade, "BloodLink Navigator System," International Journal of Research Publication and Reviews, Vol. 6, Issue 3, March 2025.
- [9] Sangeetha M., Dhivya A., Ramya R., and Sharmila N., "Blood donor searching android application," International Journal of Health Sciences, 6(S1), 13492–13503, 2021.
- [10] Ashita Jain, Amit Nirmal, Nitish Sapre, Prof. Shubhada Mone, "Online Blood Bank Management System Using Android," International Journal of Innovative Studies in Sciences and Engineering Technology (IJISSET), ISSN 2455-4863, 2016.

APPENDIX 1

RESEARCH PAPER

The research paper titled "Rakt Setu – An Integrated Mobile Platform for Expediting Blood Donation and Request Management" has been prepared for submission to a peer-reviewed journal or conference. The paper details the MERN-based architecture, functionality, and Docker deployment of Rakt Setu, along with empirical performance analysis.

Authors: Ayush Srivastava, Shriyansh Srivastava, Shaurya Jain, Prapti Sharma

Supervisor: Mr. Thammali Gangadhar

Institution: KIET Group of Institutions, Ghaziabad, Delhi-NCR, UP, India

PROOF OF SUBMISSION/ACCEPTANCE/PUBLICATION



9th International Conference on
Inventive Computation Technologies
ICICT 2026
15-17, April 2026

icict.conf@gmail.com
<https://icicts.com/2026/>

Acceptance Letter

Paper ID	Title	Author(s)
ICICT-092	An Integrated Mobile Platform for Expediting Blood Donation and Request Management	Ayush Srivastava, Shaurya Jain, Prapti Sharma, Shriyansh Srivastava, Thammali Gangadhar

Dear Author(s),

We are pleased to inform you that your paper has been accepted for presentation and publication at the 9th International Conference on Inventive Computation Technologies (ICICT 2026), to be held in Lalitpur, Nepal, during April 15–17, 2026. Your manuscript was reviewed by subject experts, and based on the reviewers' recommendations, it has been selected for inclusion in the conference proceedings. The accepted papers will be submitted for publication in IEEE Xplore, subject to IEEE compliance requirements. Authors are requested to complete the final submission and registration process within the prescribed deadlines.

Conference Publication Details:

ICICT ISBN: IEEE XPLORE COMPLIANT ISBN: 979-8-3315-6917-4

IEEE DVD ISBN: 979-8-3315-6916-7

IEEE Conference ID: #68280

We congratulate you on your acceptance and look forward to your participation in ICICT 2026.

Best Regards,

Dr. Subarna Shakya
Conference Chair
ICICT 2026



PLAGIARISM REPORT

Rakt-Setu-2.1-Updated.pdf

 Delhi Technological University

Document Details

Submission ID

trn:oid::27535:129316835

Submission Date

Feb 26, 2026, 10:38 PM GMT+5:30

Download Date

Feb 26, 2026, 10:38 PM GMT+5:30

File Name

Rakt-Setu-2.1-Updated.pdf

File Size

498.2 KB

7 Pages

4,655 Words

26,739 Characters



Page 1 of 11 - Cover Page

Submission ID trn:oid::27535:129316835



Page 2 of 11 - Integrity Overview

Submission ID trn:oid::27535:129316835

5% Overall Similarity

The combined total of all matches, including overlapping sources, for each database.

Filtered from the Report

- ▶ Bibliography
- ▶ Small Matches (less than 8 words)

Match Groups

-  **17 Not Cited or Quoted 5%**
Matches with neither in-text citation nor quotation marks
-  **0 Missing Quotations 0%**
Matches that are still very similar to source material
-  **0 Missing Citation 0%**
Matches that have quotation marks, but no in-text citation
-  **0 Cited and Quoted 0%**
Matches with in-text citation present, but no quotation marks

Top Sources

- 3%  Internet sources
- 3%  Publications
- 3%  Submitted works (Student Papers)

Integrity Flags

0 Integrity Flags for Review

No suspicious text manipulations found.

Our system's algorithms look deeply at a document for any inconsistencies that would set it apart from a normal submission. If we notice something strange, we'll flag it for you to review.

A Flag is not necessarily an indicator of a problem. However, we'd recommend you focus your attention there for further review.



Rakt-Setu-2.1-Updated.pdf

 Delhi Technological University

Document Details

Submission ID
trn:oid::27535:129316835

Submission Date
Feb 26, 2026, 10:38 PM GMT+5:30

Download Date
Feb 26, 2026, 10:39 PM GMT+5:30

File Name
Rakt-Setu-2.1-Updated.pdf

File Size
498.2 KB

7 Pages
4,655 Words
26,739 Characters



Page 1 of 9 - Cover Page

Submission ID trn:oid::27535:129316835



Page 2 of 9 - AI Writing Overview

Submission ID trn:oid::27535:129316835

*% detected as AI

AI detection includes the possibility of false positives. Although some text in this submission is likely AI generated, scores below the 20% threshold are not surfaced because they have a higher likelihood of false positives.

Caution: Review required.

It is essential to understand the limitations of AI detection before making decisions about a student's work. We encourage you to learn more about Turnitin's AI detection capabilities before using the tool.

Disclaimer

Our AI writing assessment is designed to help educators identify text that might be prepared by a generative AI tool. Our AI writing assessment may not always be accurate (i.e., our AI models may produce either false positive results or false negative results), so it should not be used as the sole basis for adverse actions against a student. It takes further scrutiny and human judgment in conjunction with an organization's application of its specific academic policies to determine whether any academic misconduct has occurred.

Frequently Asked Questions

How should I interpret Turnitin's AI writing percentage and false positives?

The percentage shown in the AI writing report is the amount of qualifying text within the submission that Turnitin's AI writing detection model determines was either likely AI-generated text from a large-language model or likely AI-generated text that was likely revised using an AI paraphrase tool or word spinner.

False positives (incorrectly flagging human-written text as AI-generated) are a possibility in AI models.

AI detection scores under 20%, which we do not surface in new reports, have a higher likelihood of false positives. To reduce the likelihood of misinterpretation, no score or highlights are attributed and are indicated with an asterisk in the report (*%).

The AI writing percentage should not be the sole basis to determine whether misconduct has occurred. The reviewer/instructor should use the percentage as a means to start a formative conversation with their student and/or use it to examine the submitted assignment in accordance with their school's policies.

What does 'qualifying text' mean?

Our model only processes qualifying text in the form of long-form writing. Long-form writing means individual sentences contained in paragraphs that make up a longer piece of written work, such as an essay, a dissertation, or an article, etc. Qualifying text that has been determined to be likely AI-generated will be highlighted in cyan in the submission, and likely AI-generated and then likely AI-paraphrased will be highlighted purple.

Non-qualifying text, such as bullet points, annotated bibliographies, etc., will not be processed and can create disparity between the submission highlights and the percentage shown.

